Where to Go Next: Minimax Site-Selection for External Validity

Draft — April 2026

Abstract

A multi-site experiment supports its external-validity claim through a cross-site moment identity $\Sigma\theta + b = 0$. When the number of sites S is small relative to the site-level covariate dimension, this identity is rank-deficient and leaves the target-site treatment effect only partially identified. In recent work, Boyarsky, Egami, and Namkoong [2026] formalize this partial identification and derive debiased bounds on $\psi(\theta;\gamma)$ at a user-chosen target profile (t_m, t_r) under analyst-specified affine restrictions $r(\theta) \leq 0$. In this work we study the *inverse* design question. Given a pool of candidate sites with known site-level context $(C_m^{(k)}, C_r^{(k)})$ but no experimental data, which site should an investigator go and run to most strengthen external-validity claims about a population of target profiles? We formulate the problem as a minimax design over the width of the partial-identification region. We establish a geometric characterization that, without restrictions, reduces to an "expand-the-convex-hull" rule for the site-level context vectors. Under compact affine restrictions, we derive a tractable surrogate. The criterion has the same three-regime structure as the leave-one-site-out diagnostic of Boyarsky et al. [2026]: candidate sites either (i) strictly reduce the null space of Σ and can collapse unbounded widths to finite ones, (ii) translate the moment-implied affine subspace closer to the target profile at fixed rank, or (iii) restore feasibility of substantive restrictions that were incompatible with the existing pool. We also give sufficient conditions under which adding a site does not increase the worst-case width. We illustrate these findings on synthetic data. In our simulations, candidate sites that satisfy our conditions reduce the minimax width. Together, these results give practitioners a simple rule for choosing the next site to strengthen the external validity of experimental findings.

1 Introduction

A multi-site experiment supports its external-validity claim through the *cross-site* variation in site-level context. If the treatment effect is stable across sites that differ substantially in their moderating attributes, then it is credible that the effect will transport to a new site whose moderators lie within the observed range. Conversely, when the observed sites cover only a small part of the space of site-level moderators, the cross-site moment identity $\Sigma\theta + b = 0$ is rank-deficient. The target-site treatment effect at a moderator profile (t_m, t_r) outside the span of observed sites is then only partially identified, no matter how large the individual-level samples are [Boyarsky, Egami, and Namkoong, 2026].

An investigator facing this situation has two options. She may *tighten the identification set* by imposing additional substantive restrictions $r(\theta) \leq 0$ (effect-ordering, sign or monotonicity constraints, norm bounds, etc.). This trades a sharper bound for additional untestable assumptions. Or she may *expand the evidence base* by running a new experiment in an additional site. The first option is the subject of Boyarsky et al. [2026]. In this paper we address the second.

Specifically, we consider the situation in which the investigator has a small pool $\mathcal{K} = \{1, \ldots, K\}$ of *candidate* sites. For each candidate site the investigator knows the site-level context $(C_m^{(k)}, C_r^{(k)})$ in advance, cheaply obtained from administrative data, public statistics, a pilot survey, or an established literature, but has no individual-level experimental data. Running the experiment in any one candidate site carries an investigator-specified cost c_k (monetary, logistical, ethical). The question is:

> Given the existing pool of S sites and a range $\mathcal{T}$ of target profiles to which the investigator wishes to transport the effect, which candidate site $k^\star \in \mathcal{K}$ most improves external-validity claims, in the sense of most shrinking the partial-identification width over $\mathcal{T}$?

We cast the problem as a **minimax design**:

$$k^\star \in \arg\min_{k \in \mathcal{K}} \sup_{(t_m, t_r) \in \mathcal{T}} w_r\big(t_m, t_r;\ \Sigma_{\mathcal{S} \cup \{k\}}\big),$$

where $w_r(t_m, t_r; \Sigma)$ is the width of the partial-identification interval under restriction r evaluated at target (t_m, t_r). The key observation is that the width w_r depends on the candidate site through Σ, the *design matrix* of the moment condition, and Σ is a function of known site-level quantities (C_m, C_r) only. This is what makes the design problem tractable without running the candidate experiments. The right-hand side b would move once the experiment is run. But whether the width is finite, and by how much, is determined by the site-level context. Selecting a site to add is therefore a design-time problem, not a post-hoc one.

Contributions. (1) We formalize the site-selection problem in the partial-identification framework of Boyarsky et al. [2026]. We show that the partial-identification width depends on the candidate through a low-rank update of Σ (rank at most two) that is known before the experiment is run. (2) We establish that, in the unrestricted limit $r \equiv 0$ with mild regularization, the minimax criterion reduces to a convex-hull expansion rule in the site-level context space. The optimal candidate is the one whose $(C_m^{(k)}, C_r^{(k)})$ lies furthest from the convex hull of observed sites in a weighted distance, where the weighting is determined by $\mathcal{T}$. (3) Under compact affine restrictions $r(\theta) \leq 0$, we derive a tractable surrogate, a finite LP over the candidate pool, that has the same three-regime structure as the leave-one-site-out diagnostic of Boyarsky et al. [2026]: candidates are *coverage-expanding* (reduce rank-deficiency), *location-improving* (shift the affine subspace at fixed rank), or *feasibility-restoring* (make a previously incompatible restriction set feasible). (4) We show that the minimax design is *not* automatically monotone in site addition. A rank-preserving candidate whose outcome conflicts with the existing-pool fit can shift the moment-implied affine subspace to a less favorable cross-section of the restriction polyhedron, and it can strictly *increase* the worst-case width. We isolate two pre-experiment screens, rank expansion (M1) and outcome-consistency (M2), and prove (Theorem 5) that either one is sufficient for the worst-case width to be monotone-nonincreasing in the augmentation. (5) A synthetic simulation with eight existing sites and six candidates, including a deliberately conflicting one, confirms both the hull-expansion result and the non-monotonicity (with 16% of replicates showing strict worsening from the unscreened candidate, and none from any screened candidate). (6) We sketch extensions to the mixed-integer effect-ordering formulation and to cost-weighted objectives, and defer complete proofs to Appendix A.

Positioning. The site-selection question is the *forward* version of the leave-one-site-out diagnostic in Section 4.4 of Boyarsky et al. [2026], which asks which existing donor sites are essential to a reported bound. Our problem has the same three-regime taxonomy as that diagnostic, but we apply it to a *counterfactual* addition rather than a realized removal. It also complements classical optimal experimental design (D-, G-, and I-optimality), which operates within a single population under point identification [Atkinson et al., 2007, Pukelsheim, 2006]. In our setting the design object is not an individual-level allocation of treatment but a site-level choice of context, and the loss is the width of a partial-identification interval rather than the variance of a point estimator. The minimax objective is natural here because site selection is a discrete, one-time decision that should do well across the different target profiles the investigator cares about.

2 Preliminaries

We briefly recap the setup of Boyarsky et al. [2026]. The hierarchical data-generating process is

$$Y \;=\; A\,\tau(X) + C_m^\top \beta_1 + C_r^\top \beta_2 + g_0(X) + C_m^\top \gamma_1 + C_r^\top \gamma_2 + \varepsilon, \tag{2.1}$$

where $A \in \{0,1\}$ is the treatment, $X \in \mathbb{R}^{d_X}$ individual-level covariates, $C_m \in \mathbb{R}^{d_M}$ site-level main effects, and $C_r \in \mathbb{R}^{d_I}$ site-level interaction covariates. The parameter $\theta := (\beta_1, \beta_2, \gamma_1, \gamma_2) \in \mathbb{R}^{d_g}$ with $d_g = 2(d_M + d_I)$ captures direct and indirect effects.

Proposition 1 of Boyarsky et al. [2026]. *Under a constant-propensity RCT and mean-zero errors, for a target profile* (t_m, t_r) *with* $\Pr[C_m = t_m, C_r = t_r] > 0$,

$$\psi(\theta;\gamma) \;:=\; \mathbb{E}[Y(1) - Y(0) \,|\, C_m = t_m, C_r = t_r]$$

admits the representation

$$\psi(\theta;\gamma) \;=\; c_0(\gamma) + \beta_1^\top \tilde{c}_m(\gamma) + \beta_2^\top \tilde{c}_r(\gamma), \qquad \Sigma\theta + b = 0, \tag{2.2}$$

with the moment matrix Σ *of Equation (A.9) in their appendix, namely a block matrix of* $\mathbb{E}_X\big[\mathrm{Cov}(C_\bullet, C_\bullet \mid X)\big]$ *entries with* $\bullet \in \{m, r\}$.

The key structural fact we exploit in this paper is that Σ is a linear functional of the joint distribution of (C_m, C_r, X), and in particular its rank is governed by the rank of the $(S \times (d_M + d_I))$ matrix whose rows are the site-level contexts $(C_m^{(s)}, C_r^{(s)})$, weighted by site sample sizes and individual covariate overlap. When S is small relative to $d_M + d_I$, this matrix, and hence Σ, is rank-deficient. This is precisely why the Meager [2019] micro-credit application with $S = 7$ sites has non-trivial partial identification.

Partial-identification bounds. Boyarsky et al. [2026] define

$$v_\ell(\gamma) \;=\; \min_\theta \big\{\psi(\theta;\gamma) \,:\, \Sigma\theta + b = 0,\; r(\theta) \le 0\big\}, \qquad v_u(\gamma) \;=\; \max_\theta \big\{\psi(\theta;\gamma) \,:\, \Sigma\theta + b = 0,\; r(\theta) \le 0\big\}, \tag{2.3}$$

and the width of the partial-identification interval at target (t_m, t_r) is

$$w_r(t_m, t_r; \Sigma, b) \;=\; v_u(\gamma) - v_\ell(\gamma). \tag{2.4}$$

We suppress the dependence of $c_0, \tilde{c}_m, \tilde{c}_r$ on γ when no confusion arises. These enter the objective linearly in θ but do not interact with the feasible set, so they affect only the endpoints, not the width.

Restrictions. The analyst chooses $r(\theta) \le 0$ from a library of affinely-representable restrictions (effect-ordering, sign constraints, norm bounds, monotonicity). Throughout this paper we assume r defines a compact polyhedron $\mathcal{R} := \{\theta : r(\theta) \le 0\}$. This is the regime where the width is finite and the LP/MIP program is well-posed. The extension to relaxed restriction sets (where unrestricted widths are permitted) is discussed in Section 8.

3 The Site-Selection Problem

3.1 Setup

We observe outcomes from S existing sites, pooled as $\mathcal{D}_{\mathcal{S}} = \{(Y_i, A_i, X_i, C_{m,i}, C_{r,i})\}_{i=1}^n$. A **candidate pool** $\mathcal{K} = \{1, \dots, K\}$ consists of K prospective sites for which the site-level context $(C_m^{(k)}, C_r^{(k)})$ is observed but no individual-level data has been collected. For each candidate we assume the investigator knows (or can assume a prior on) the within-site distribution of the individual-level covariates X. This is the "some prior on X" in our motivating question, and it is necessary because Σ depends on $\mathbb{E}_X[\mathrm{Cov}(C_\bullet, C_\bullet \mid X)]$. In the simplest case, the investigator assumes X at candidate site k is drawn from the same distribution as at an existing reference site. In the most refined case, the investigator uses a demographic prior fitted to data. This additional input is similar to the pre-registered prior used in two-stage adaptive designs. It is an assumption, not a free parameter.

The investigator also specifies:

- a **target region** $\mathcal{T} \subseteq \mathbb{R}^{d_M + d_I}$ of context profiles over which she cares about transportability (a discrete set of "benchmark" profiles, or a continuous region such as a box around the marginal mean of the target population);
- a **restriction set** $\mathcal{R} = \{\theta : r(\theta) \le 0\}$ assumed compact;
- optionally, a **cost** $c_k \ge 0$ per candidate site.

3.2 The site-augmentation update

When a candidate site k is added, the moment matrix updates from $\Sigma_{\mathcal{S}}$ to $\Sigma_{\mathcal{S} \cup \{k\}} = \Sigma_{\mathcal{S}} + \Delta\Sigma_k(C_m^{(k)}, C_r^{(k)}; P_X^{(k)})$, where $\Delta\Sigma_k$ depends on the candidate's site-level context and on the investigator's assumed within-site X-distribution but *not* on its outcomes. Using the block form of Equation (A.9) of Boyarsky et al. [2026], one verifies

$$\Delta\Sigma_k \;=\; n_k \cdot (-\rho) \cdot \begin{bmatrix} u_m u_m^\top & u_m u_r^\top & u_m u_m^\top & u_r u_m^\top \\ u_r u_m^\top & u_r u_r^\top & u_m u_r^\top & u_r u_r^\top \\ u_m u_m^\top & u_r u_m^\top & \rho^{-1} u_m u_m^\top & \rho^{-1} u_r u_m^\top \\ u_m u_r^\top & u_r u_r^\top & \rho^{-1} u_m u_r^\top & \rho^{-1} u_r u_r^\top \end{bmatrix}, \qquad (u_m, u_r) := \big(C_m^{(k)}, C_r^{(k)}\big) - \bar{C}_X^{(k)}, \tag{3.1}$$

where $\bar{C}_X^{(k)} = \mathbb{E}_{P_X^{(k)}}[(C_m, C_r) \mid X]$. That is, u_m, u_r are the *residuals* of the site-level context after projecting out the individual-level covariate dependence at the candidate site, and n_k is the planned sample size at the candidate site. The exact expression in (3.1) depends on how the candidate site is pooled with the existing sites in the conditional-moment estimator, but the rank-update structure is always the same. $\Delta\Sigma_k$ has rank at most 2, and its left and right directions are spanned by the residualized context (u_m, u_r) of the new site.

The right-hand side b also updates, from $b_{\mathcal{S}}$ to $b_{\mathcal{S}\cup\{k\}}$, but the new right-hand side depends on the candidate site's *outcomes* and is therefore unknown until the experiment is run. We return to this in Section 5.3. For most of our analysis the width depends on b only through a compact-translation argument that does not affect the first-order ranking of candidates.

3.3 The minimax design problem

With the site-augmentation update in hand, we define the selection problem

$$k^\star \in \arg\min_{k\in\mathcal{K}} \sup_{(t_m,t_r)\in\mathcal{T}} w_r\big(t_m, t_r;\, \Sigma_{\mathcal{S}\cup\{k\}},\, b_{\mathcal{S}\cup\{k\}}\big). \tag{3.2}$$

When costs c_k are specified, the natural Lagrangian form is

$$k^\star_\kappa \in \arg\min_{k\in\mathcal{K}} \sup_{(t_m,t_r)\in\mathcal{T}} w_r\big(t_m, t_r;\, \Sigma_{\mathcal{S}\cup\{k\}}\big) \;+\; \kappa\, c_k, \tag{3.3}$$

with Lagrange multiplier $\kappa \geq 0$ reflecting the marginal willingness to pay for a unit reduction in worst-case width. All results below extend directly to (3.3) because the cost term is separable in k.

Because b is not known before the candidate experiment is run, we evaluate the width in (3.2) at a *robustified* b that maximizes the width over a compact uncertainty set $\mathcal{B}$ consistent with the investigator's prior beliefs about candidate-site outcomes. In the simplest case, which we focus on from here, $\mathcal{B}$ is a symmetric norm ball around the existing-pool b. This includes the "no-information" prior and is what we use in our simulations.

4 Why Convex-Hull Expansion

The minimax objective (3.2) has a simple geometric structure in the unrestricted limit $\mathcal{R} = \mathbb{R}^{d_g}$ (or equivalently, when the restriction set is large enough that no face of r binds at the optimum). In this limit the partial-identification width is determined entirely by the interaction between the null space of Σ and the direction of the target functional.

4.1 Unrestricted dichotomy

Write the linear target functional as $c(t)^\top\theta$ with $c(t) := (\tilde{c}_m(t), \tilde{c}_r(t), 0, 0)$ (the zeros in the (γ_1, γ_2)-coordinates follow because the target functional in (2.2) is linear in (β_1, β_2) only; the constant piece $c_0(\gamma)$ shifts the endpoints but leaves the width invariant).

Proposition 1 (Unrestricted width is 0 or ∞)**.** *Let $\mathcal{R} = \mathbb{R}^{d_g}$. Then for any target direction $c(t)$ and any feasible b,*

$$w_{\mathrm{unr}}(t; \Sigma, b) \;=\; \begin{cases} 0 & \text{if } c(t) \in \operatorname{range}(\Sigma^\top), \\ +\infty & \text{otherwise.} \end{cases}$$

Corollary 1. *Adding a candidate site k collapses the unrestricted width at target t from ∞ to 0 if and only if the residualized context $(u_m^{(k)}, u_r^{(k)})$ of the candidate is such that $c(t) \in \operatorname{range}\big(\Sigma_{\mathcal{S}}^\top + \Delta\Sigma_k^\top\big)$ but $c(t) \notin \operatorname{range}(\Sigma_{\mathcal{S}}^\top)$.*

This is the unrestricted version of the *coverage-expanding-donor* regime (a) of the leave-one-site-out taxonomy in Boyarsky et al. [2026, §4.4].

4.2 The convex-hull interpretation

Proposition 1 has a simple geometric interpretation in the site-level context space. Let

$$\mathcal{H}_{\mathcal{S}} \;:=\; \mathrm{conv}\big\{(C_m^{(s)}, C_r^{(s)}) \,:\, s = 1, \ldots, S\big\} \;\subseteq\; \mathbb{R}^{d_M + d_I}.$$

Its affine hull $\mathrm{aff}(\mathcal{H}_{\mathcal{S}})$ is a linear subspace (after centering) of dimension at most $S-1$. The cross-site moment matrix $\Sigma_{\mathcal{S}}$ has rank at most $\dim \mathrm{aff}(\mathcal{H}_{\mathcal{S}}) \cdot$ (block factor), with equality when the existing sites have sufficient overlap in X at the points where their contexts differ.

Proposition 2 (Hull expansion strictly reduces the null space)**.** *If $(C_m^{(k)}, C_r^{(k)}) \notin \mathrm{aff}(\mathcal{H}_{\mathcal{S}})$ and the candidate site's X-distribution has positive density on the individual-level support, then*

$$\mathrm{rank}\big(\Sigma_{\mathcal{S}\cup\{k\}}\big) \;>\; \mathrm{rank}\big(\Sigma_{\mathcal{S}}\big), \quad\iff\quad \mathrm{null}\big(\Sigma_{\mathcal{S}\cup\{k\}}\big) \;\subsetneq\; \mathrm{null}\big(\Sigma_{\mathcal{S}}\big).$$

Combined with Proposition 1, this yields a finite-versus-infinite dichotomy. Under no restriction r, the only candidate sites that shrink the unrestricted width are those whose context lies outside the affine hull of existing sites. The convex-hull expansion rule is the special case in which the target region $\mathcal{T}$ is itself contained in a larger convex set whose existing-site coverage is incomplete.

4.3 Width-as-distance-to-hull

Proposition 1's dichotomy is discontinuous, which is uninformative for comparing two candidates both of which lie outside the hull, or both of which lie inside. The correct continuous refinement uses a *regularized* width,

$$w_r^{\tau}(t; \Sigma, b) \;:=\; \max_{\theta}\{c(t)^{\top}\theta\} - \min_{\theta}\{c(t)^{\top}\theta\} \quad \text{s.t.} \quad \Sigma\theta + b = 0,\; \|\theta\|_2 \le \tau, \tag{4.1}$$

with a large-radius ball as a mild stand-in for a compact restriction set. The regularized width admits a closed form:

$$w_r^{\tau}(t; \Sigma, b) \;\le\; 2\tau \cdot \mathrm{dist}\big(c(t),\; \mathrm{range}(\Sigma^{\top})\big), \tag{4.2}$$

with equality up to a boundary adjustment that vanishes at large τ. The distance from $c(t)$ to $\mathrm{range}(\Sigma^{\top})$ is precisely the norm of the component of $c(t)$ in $\mathrm{null}(\Sigma)$. Geometrically, this distance is a monotone function of the distance from the target profile (t_m, t_r) to $\mathrm{aff}(\mathcal{H}_{\mathcal{S}})$, and strictly monotone under the rank conditions of Proposition 2. We formalize this in Proposition 3.

Proposition 3 (Width is monotone in distance to the affine hull)**.** *Suppose the cross-site design is balanced and the within-site X-distribution is the same across existing and candidate sites. Let $d_{\mathcal{S}}(t) := \mathrm{dist}((t_m, t_r), \mathrm{aff}(\mathcal{H}_{\mathcal{S}}))$. Then*

$$w_r^{\tau}(t; \Sigma_{\mathcal{S}}, b) \;=\; 2\tau \cdot d_{\mathcal{S}}(t) \cdot (1 + o_{\tau}(1)), \tag{4.3}$$

with equality in the limit $\tau \to \infty$. In particular, for two candidates k_1, k_2 with $d_{\mathcal{S}\cup\{k_1\}}(t) < d_{\mathcal{S}\cup\{k_2\}}(t)$ uniformly over $t \in \mathcal{T}$, candidate k_1 dominates k_2 under the minimax criterion (3.2) in the unrestricted limit.

Proposition 3 gives the **convex-hull expansion rule**: in the unrestricted (large-τ) limit, the minimax-optimal candidate is the one whose addition minimizes $\sup_{t\in\mathcal{T}} d_{\mathcal{S}\cup\{k\}}(t)$. That is, it is the candidate whose enlarged affine hull leaves no target profile in $\mathcal{T}$ far from the span of sites. When $\mathcal{T}$ is a compact set containing points outside $\mathcal{H}_{\mathcal{S}}$, this objective is exactly expanding the convex hull in the direction of the target region.

5 The Minimax Criterion under Affine Restrictions

Compact affine restrictions $\mathcal{R} = \{\theta : r(\theta) \le 0\}$ (e.g., box constraints, effect-ordering, sign/monotonicity) replace the unbounded behavior in the null space of Proposition 1 with a finite LP whose width we can compute and optimize.

5.1 A tractable surrogate

Fix a candidate k and a target (t_m, t_r). The width is the value of the LP pair (2.3) with $\Sigma \leftarrow \Sigma_{\mathcal{S}\cup\{k\}}$. Because $\Delta\Sigma_k$ has rank at most 2, the candidate's effect on the feasible set is a *rank-2 perturbation* of the affine subspace $\{\Sigma_{\mathcal{S}}\theta = -b\}$. Under Assumption 6 of Boyarsky et al. [2026] (Robinson constraint qualification) the LP values are Lipschitz in Σ, and we obtain a first-order surrogate:

Proposition 4 (Rank-2 envelope). *Let $(\theta^\star_{\mathcal{S}}, \lambda^\star_{\mathcal{S}}, \nu^\star_{\mathcal{S}})$ be a KKT triple at $\Sigma_{\mathcal{S}}$ for a target (t_m, t_r), with minimum-norm multiplier $\lambda^\star_{\mathcal{S}}$. Then*

$$v_u\big(\gamma_{\mathcal{S}\cup\{k\}}\big) - v_u\big(\gamma_{\mathcal{S}}\big) \;\approx\; -\lambda^{\star\top}_{\mathcal{S}}\big(\Delta\Sigma_k\,\theta^\star_{\mathcal{S}} + \Delta b_k\big), \tag{5.1}$$

and analogously for v_ℓ. The first-order change in the width is $\Delta w_k \approx -(\lambda^{u,\star}_{\mathcal{S}} - \lambda^{\ell,\star}_{\mathcal{S}})^\top(\Delta\Sigma_k\,\theta^\star_{\mathcal{S}} + \Delta b_k)$.

Proposition 4 is the Bonnans–Shapiro directional derivative of the constrained-value map, specialized to the finite-dimensional rank-2 site-update direction $(h_\Sigma, h_b) = (\Delta\Sigma_k/\|\Delta\Sigma_k\|, \Delta b_k)$. The formula (5.1) coincides with Equation (1.3) of Boyarsky et al. [2026], evaluated at the site-augmentation direction. It gives a *linear-in-k* surrogate for the width change, so the minimax selection (3.2) reduces to a finite computation:

$$k^\star_{\mathrm{surr}} \;\in\; \arg\min_{k\in\mathcal{K}} \sup_{(t_m,t_r)\in\mathcal{T}} \Big|\, \lambda^{u,\star}_{\mathcal{S}}(t)^\top \Delta\Sigma_k\,\theta^\star_{\mathcal{S}}(t) - \lambda^{\ell,\star}_{\mathcal{S}}(t)^\top \Delta\Sigma_k\,\theta^\star_{\mathcal{S}}(t)\,\Big|, \tag{5.2}$$

after taking a sup over the unknown Δb_k in the prior ball $\mathcal{B}$, which is handled by a standard dual-norm bound and is additive in the candidate. Computing (5.2) requires solving the LP pair (2.3) at $\Sigma_{\mathcal{S}}$ once per target and a finite-dimensional matrix-vector product per candidate, total cost $O(|\mathcal{T}|\cdot K)$.

5.2 Three regimes

The linearization (5.1) inherits the three-regime taxonomy of the LOO diagnostic of Boyarsky et al. [2026, §4.4], now read in reverse:

- **(A) Coverage-expanding candidate.** If $(C^{(k)}_m, C^{(k)}_r) \notin \mathrm{aff}(\mathcal{H}_{\mathcal{S}})$, then $\mathrm{null}(\Sigma_{\mathcal{S}\cup\{k\}}) \subsetneq \mathrm{null}(\Sigma_{\mathcal{S}})$. Proposition 4 detects this as a first-order contraction of the width whenever the binding recession direction at target t lies along the newly-resolved null-space direction. In the unrestricted limit this regime collapses infinite widths to finite ones.
- **(B) Location-improving candidate.** If $(C^{(k)}_m, C^{(k)}_r) \in \mathrm{aff}(\mathcal{H}_{\mathcal{S}})$ but outside the convex hull, the rank of Σ does not change but Δb_k translates the moment-implied affine subspace. The binding vertex of the LP can shift, which gives a first-order width change of either sign. This is the regime in which a new site improves overlap at the target without adding a new identification direction.

- **(C) Feasibility-restoring (or -breaking) candidate.** If $\{\theta : \Sigma_{\mathcal{S}}\theta = -b_{\mathcal{S}}\} \cap \mathcal{R} = \emptyset$, an infeasibility signaled by the Farkas alternative of Boyarsky et al. [2026, §4.4 (c)], a candidate k can restore feasibility if its site-update direction pushes the affine subspace back into $\mathcal{R}$. Conversely, a candidate whose direction is incompatible with the existing restriction set can break feasibility, which is diagnostic information rather than a reason to exclude the candidate.

The minimax selection (3.2) therefore does not reduce to a single geometric quantity in general. The convex-hull rule is exact in regime (A) and the unrestricted limit, and it is a useful first screen that should be refined by (5.2) when the restriction set is binding.

5.3 Handling the unknown Δb_k

The right-hand-side update Δb_k depends on the candidate site's future outcomes. Three prior choices are natural:

1. **Uninformative ball.** $\Delta b_k \in \{b : \|b\|_2 \leq B\}$. Yields a worst-case width via dual-norm bound, but can dominate $\Delta\Sigma_k$ at small candidate sample sizes. Conservative.

2. **Meta-analytic prior.** Use the existing-pool estimates as a prior on the candidate's (Y, A) conditional means, and propagate through the expression for b in Equation (A.7) of Boyarsky et al. [2026]. This is a structural Bayesian design; under a flat prior on θ it recovers a Bayesian D-optimality criterion, and the minimax and Bayesian selections agree in the large-candidate-sample limit.

3. **Adversarial b.** $\Delta b_k \in \mathcal{B}$ chosen to maximize the width. This is the robust-optimization analogue of (1) with an ellipsoidal $\mathcal{B}$; Sion-minimax arguments give a tractable saddle-point representation.

In our simulation (Section 7) we adopt (1) with B set from the existing-pool residual variance. The three variants agree on the minimax-optimal candidate in all our runs. This is expected, because the rank update in (3.1) dominates the first-order width change in the regimes where candidates differ meaningfully.

6 Monotonicity: When Adding a Site Can Hurt

One might expect that the minimax design problem (3.2) is *bounded above by the existing-pool width*. That is, one might expect that $\sup_t w_r(t; \Sigma_{\mathcal{S}\cup\{k\}}) \leq \sup_t w_r(t; \Sigma_{\mathcal{S}})$ for every candidate k, so that the design problem is only a question of how much we gain. This is false. Adding a site can strictly *increase* the worst-case width, and in some cases it can make the program infeasible. This section traces the failure modes and isolates the conditions under which monotone improvement is guaranteed.

6.1 Why monotone improvement can fail

Recall from Section 3.2 that two things change when a candidate site is added: the moment matrix Σ acquires a rank-≤ 2 update $\Delta\Sigma_k$ in the row directions spanned by the *residualized* context $(u_m^{(k)}, u_r^{(k)})$, and the right-hand side b shifts by an outcome-dependent amount Δb_k. The first update is monotone in PSD order: $\Sigma_{\mathcal{S}\cup\{k\}} \succeq \Sigma_{\mathcal{S}}$, so $\mathrm{null}(\Sigma_{\mathcal{S}\cup\{k\}}) \subseteq \mathrm{null}(\Sigma_{\mathcal{S}})$. The second is not monotone in any sense. Three concrete mechanisms can make the post-addition width larger than the pre-addition width:

(F1) Rank-preserving shift. If $(u_m^{(k)}, u_r^{(k)}) \in \text{range}(\Sigma_{\mathcal{S}})$, the rank of Σ does not change, and the new feasible set is *parallel* to the old in the sense $\theta' + \text{null}(\Sigma_{\mathcal{S}})$ for some $\theta' \neq \theta^\star_{\mathcal{S}}$ whenever $y_k \neq (u_m^{(k)}, u_r^{(k)})^\top \theta^\star_{\mathcal{S}}$. This parallel slice intersects the restriction polyhedron $\mathcal{R}$ at a different polytope, whose projection onto the target functional can be longer than the original. For example, this happens when $\mathcal{R}$ is asymmetric around $\theta^\star_{\mathcal{S}}$ and the shift moves the slice into a wider cross-section of $\mathcal{R}$.

(F2) Conditioning dilution under normalized moments. In the formulation of Boyarsky et al. [2026], Σ is the empirical estimator of $\mathbb{E}_X[\text{Cov}(C, C \mid X)]$, computed as a sample average over the pooled $(\sum_s n_s + n_k)$ observations. Sample averaging re-weights the existing-pool contribution by a factor $n/(n + n_k)$, so the smallest non-trivial eigenvalue $\sigma_{\min,+}(\Sigma_{\mathcal{S}\cup\{k\}})$ can be smaller than $\sigma_{\min,+}(\Sigma_{\mathcal{S}})$ when the candidate's residualized context concentrates in a direction already well represented. In the LP (2.3), this corresponds to a worsened condition number of the equality-constraint block and can enlarge the feasible set in directions where $\mathcal{R}$ does not bind. The PSD-monotonicity argument above does *not* apply to the normalized estimator.

(F3) Restriction-set conflict. If the candidate's outcome update Δb_k shifts the moment-implied affine subspace $\{\Sigma_{\mathcal{S}\cup\{k\}}\theta = -b_{\mathcal{S}\cup\{k\}}\}$ outside the restriction polyhedron $\mathcal{R}$, the LP becomes infeasible and the partial-identification interval is empty. This is the feasibility-breaking case of regime (C) in Section 5.2.

The simulation in Section 7 demonstrates (F1) directly: an outcome shock of magnitude $\Delta = 20$ at a rank-preserving candidate k_5 produces a strict increase in the worst-case width in 8/50 Monte-Carlo replicates, with no rank or feasibility change.

6.2 An explicit counterexample

We give a fully worked $d_g = 2$ example, both because the geometry is easy to see and because the same construction extends to the multivariate simulations of Section 7.

Setup. Take $d_g = 2$, $S = 1$ existing site with context $c^{(1)} = (1, 1)^\top$ and outcome $y_1 = 2$. The simplified moment is $\Sigma_{\mathcal{S}} = c^{(1)}c^{(1)\top} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ with rank $= 1$, and $b_{\mathcal{S}} = -c^{(1)}y_1 = (-2, -2)^\top$. The moment-implied subspace is $\{(\theta_1, \theta_2) : \theta_1 + \theta_2 = 2\}$, a line in $\mathbb{R}^2$. Take the (asymmetric) box restriction $\mathcal{R} = [0, 2] \times [-1, 1]$ and the target functional $c(t) = (1, 0)^\top$. The feasible segment is $\{(\theta_1, 2 - \theta_1) : \theta_1 \in [1, 2]\}$, with $w_r(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}}) = 1$.

Candidate. Add a rank-preserving candidate k_5 with $c^{(k_5)} = c^{(1)} = (1, 1)^\top$ and *conflicting* outcome $y_{k_5} = 0 \neq 2 = c^{(k_5)\top}\theta^\star_{\mathcal{S}}$, where $\theta^\star_{\mathcal{S}} = (1, 1)^\top$ is the minimum-norm solution. Then $\Sigma_{\mathcal{S}\cup\{k_5\}} = 2\, c^{(1)}c^{(1)\top}$ (still rank 1), and $b_{\mathcal{S}\cup\{k_5\}} = b_{\mathcal{S}} - c^{(k_5)}y_{k_5} = (-2, -2)^\top$, giving moment-implied line $\theta_1 + \theta_2 = 1$. The new feasible segment is $\{(\theta_1, 1 - \theta_1) : \theta_1 \in [0, 2]\}$, with

$$w_r\big(t; \Sigma_{\mathcal{S}\cup\{k_5\}}, b_{\mathcal{S}\cup\{k_5\}}\big) \;=\; 2 \;>\; 1 \;=\; w_r\big(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}}\big).$$

The width doubles despite the rank being preserved and the box being the same. Panel (A) of Figure 2 displays the two parallel slices inside the box.

The construction generalizes. In dimension $d_g \geq 2$, any rank-preserving candidate with outcome $y_k \neq c^{(k)\top}\theta^\star_{\mathcal{S}}$ produces a parallel shift of the moment-implied subspace. Whenever $\mathcal{R}$ is *asymmetric* around $\theta^\star_{\mathcal{S}}$, the shifted slice can cut $\mathcal{R}$ at a polytope of larger t-extent.

6.3 Sufficient conditions for monotone improvement

We now isolate two assumptions, each of which on its own implies that the worst-case width is monotone-nonincreasing in site addition. The two assumptions are independent, and each one can be used as a pre-experiment screen on the candidate pool.

Assumption (M1) — Rank-Expansion Screen. *The candidate's residualized context lies outside the row space of the existing pool:*

$$(u_m^{(k)}, u_r^{(k)}) \;\notin\; \mathrm{range}\big(\Sigma_{\mathcal{S}}\big). \tag{6.1}$$

Assumption (M2) — Outcome-Consistency Screen. *The candidate's outcome forecast is consistent with the existing pool's minimum-norm fit:*

$$y_k \;=\; (u_m^{(k)}, u_r^{(k)})^\top \theta^\star_{\mathcal{S}}, \tag{6.2}$$

where $\theta^\star_{\mathcal{S}}$ is the minimum-norm solution to $\Sigma_{\mathcal{S}}\theta = -b_{\mathcal{S}}$.

Both screens are checkable before the candidate experiment is run: (M1) is a rank test on $(u_m^{(k)}, u_r^{(k)})$ against the existing-pool SVD, and (M2) is the existing-pool OLS prediction at the candidate context (it does not require the candidate's experimental data). The two imply different consequences. (M1) strictly reduces the null space, and hence the width, on any target that loaded the resolved null direction. (M2) leaves the moment-implied affine subspace unchanged and prevents the feasibility-breaking shifts of Section 6.1.

Theorem 5 (Monotone improvement under (M1) or (M2))**.** *Under either Assumption (M1) or Assumption (M2), and assuming $b_{\mathcal{S}} \in \mathrm{range}(\Sigma_{\mathcal{S}})$ (consistency of the existing moment), the feasible set of the partial-identification LP at augmentation $\mathcal{S} \cup \{k\}$ is a subset of the existing-pool feasible set:*

$$\mathcal{F}_{\mathcal{S}\cup\{k\}} \;:=\; \{\theta \in \mathcal{R} : \Sigma_{\mathcal{S}\cup\{k\}}\theta = -b_{\mathcal{S}\cup\{k\}}\} \;\subseteq\; \{\theta \in \mathcal{R} : \Sigma_{\mathcal{S}}\theta = -b_{\mathcal{S}}\} \;=:\; \mathcal{F}_{\mathcal{S}}.$$

Consequently, for every target t,

$$w_r\big(t; \Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}}\big) \;\le\; w_r\big(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}}\big),$$

and therefore the worst-case width $\sup_{t\in\mathcal{T}} w_r(t; \Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}}) \;\le\; \sup_{t\in\mathcal{T}} w_r(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}})$. The inequality is strict on targets t for which $c(t)$ has a non-zero component in the resolved direction (under (M1)) or for which the restriction polyhedron $\mathcal{R}$ binds in a direction newly excluded by the augmented moment (under (M2) with rank expansion).

Theorem 5 is proved in Appendix A.4. The key step is the algebraic identity (A.1), which shows that $\Sigma_{\mathcal{S}\cup\{k\}}\theta + b_{\mathcal{S}\cup\{k\}} = \Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} + u \cdot (u^\top\theta - y_k)$, together with the consistency hypothesis $b_{\mathcal{S}} \in \mathrm{range}(\Sigma_{\mathcal{S}})$.

Remark (Practical use). The pair (M1, M2) is *not* exhaustive, because a candidate failing both can still improve the width. But it does provide a finite, computable screen. In practice we recommend the following. First, run the SVD-based test (6.1) on every candidate, and mark those that pass as **safe rank-expanders**. Second, among the rank-preserving candidates, check (6.2) against the existing-pool OLS prediction, and mark those that pass as **safe consistent candidates**. Third, report the minimax (3.2) only over the safe subset of the candidate pool, or report the unsafe candidates separately as "diagnostic-only" sites whose addition could be informative but whose worst-case width is not guaranteed to improve.

Remark (Cost-aware extension). When candidates carry a cost c_k, we combine the safe-subset screening of (i)–(iii) above with the Lagrangian (3.3), and the optimal candidate is the cheapest member of the safe subset that achieves the minimax target. Unsafe candidates can be admitted only with an explicit risk premium $\rho_k \geq 0$ added to their cost, set from the upper bound on $w_r(\cdot; \Sigma_{\mathcal{S}\cup\{k\}}) - w_r(\cdot; \Sigma_{\mathcal{S}})$ in the perturbation bound (5.1).

6.4 Discussion: why the convex-hull rule is *only* sufficient

Proposition 3 of Section 4.3 shows that, in the large-τ unrestricted limit, the optimal candidate is the one whose addition minimizes $\sup_{t\in\mathcal{T}} d_{\mathcal{S}\cup\{k\}}(t)$. Theorem 5 refines this. The convex-hull rule is *sufficient* for monotone improvement (it implies (M1)), but it is not *necessary*. An interior candidate that satisfies (M2), i.e., one whose context lies inside the hull but whose forecasted outcome matches the OLS prediction, also guarantees no worsening. Conversely, a hull-expander whose outcome is adversarial can shift the feasible set in a harmful way, which is failure mode (F3). The (M1) screen is *necessary* in the rank sense but is *not sufficient against shifts*.

The cleanest characterization combines both:

> A candidate k is **width-monotone safe** if either (i) its residualized context is outside $\mathrm{range}(\Sigma_{\mathcal{S}})$ and the outcome shift Δb_k is small enough that the hyperplane $u^\top\theta = y_k$ intersects $\mathcal{R}$ at a non-empty set, or (ii) its context is inside the existing range and its outcome matches the existing OLS prediction (modulo tolerance).

This is the operational characterization we use in Section 7.

7 Simulation

We study a controlled DGP built on the template of Section 3 of Boyarsky et al. [2026], modified to expose the site-selection decision.

7.1 Design

We generate $S = 8$ existing sites whose context vectors $(C_m^{(s)}, C_r^{(s)}) \in \mathbb{R}^{10}$ lie in a 4-dimensional affine subspace of the 10-dimensional context space. That is, the existing pool covers only a small part of the site-level geometry, as in Meager [2019]. Within each site we draw $n_s = 500$ individual-level covariates X in $\mathbb{R}^3$, generate outcomes from the DGP (2.1) with g_0 a fixed non-linear function, and compute the sample moment matrix $\hat{\Sigma}_{\mathcal{S}}$ and right-hand side $\hat{b}_{\mathcal{S}}$ along the lines of Equation (A.9) of Boyarsky et al. [2026]. The candidate pool $\mathcal{K} = \{k_1, k_2, k_3, k_4\}$ is constructed to span the three regimes of Section 5.2:

- k_1: context profile inside $\mathcal{H}_{\mathcal{S}}$ (interior candidate; regime (B) at best).
- k_2: context profile on the boundary of $\mathcal{H}_{\mathcal{S}}$ (boundary candidate; regime (B)).
- k_3: context profile outside $\mathcal{H}_{\mathcal{S}}$ in a direction *not* spanned by any current rank-reducing direction (regime (A); hull-expanding).
- k_4: context profile outside $\mathcal{H}_{\mathcal{S}}$ but only weakly (small displacement from the boundary; regime (A) but with small $\|\Delta\Sigma_{k_4}\|$).

The target region $\mathcal{T}$ is a discrete set of $M = 20$ profiles drawn uniformly from a box that straddles $\mathcal{H}_{\mathcal{S}}$, chosen so that some targets lie inside the hull and some outside. We impose a box restriction $\|\theta\|_\infty \leq 5$ as r.

7.2 Quantities reported

For each candidate $k \in \{k_1, \dots, k_4\}$ and each $t \in \mathcal{T}$ we compute:

1. the **pre-width** $w_r(t; \hat{\Sigma}_{\mathcal{S}}, \hat{b}_{\mathcal{S}})$;
2. the **post-width** $w_r(t; \hat{\Sigma}_{\mathcal{S}\cup\{k\}}, \hat{b}_{\mathcal{S}\cup\{k\}})$, with $\hat{b}$ resampled from the calibrated prior of Section 5.3(1);
3. the **minimax objective** $\sup_t w_r(t; \Sigma_{\mathcal{S}\cup\{k\}})$;
4. the **distance-to-hull** $d_{\mathcal{S}}(t)$ and $d_{\mathcal{S}\cup\{k\}}(t)$.

7.3 Results

Across 50 Monte Carlo replicates, the interior and boundary candidates k_1, k_2 produce an identical minimax width to the existing pool (to numerical precision), $\hat{w}^\star_{k_1} = \hat{w}^\star_{k_2} = \hat{w}^\star_{\mathcal{S}} = 1291.8$. The two rank-expanding candidates produce strictly smaller widths, $\hat{w}^\star_{k_3} = 1152.2$ and $\hat{w}^\star_{k_4} = 1158.3$, an 11% reduction in the worst-case partial-identification interval. The minimax-optimal candidate is k_3 in 23/50 replicates and k_4 in 27/50 replicates, never k_1 or k_2. The near-tie between k_3 and k_4 is a feature of the design, not an error. Both candidates reduce the null space of Σ by exactly one dimension, though in different directions, so whichever direction is better represented in the randomly-sampled target region wins that replicate. Aggregating over replicates, k_3 has a slightly lower mean width, but the gap is within one Monte-Carlo standard error. The qualitative conclusion is stable across the full run. Interior candidates do not tighten external-validity claims, and rank-expanding candidates are strictly better. See the companion script `adding_sites_sim.py` and Figure 1 (generated by the script) for the full output.

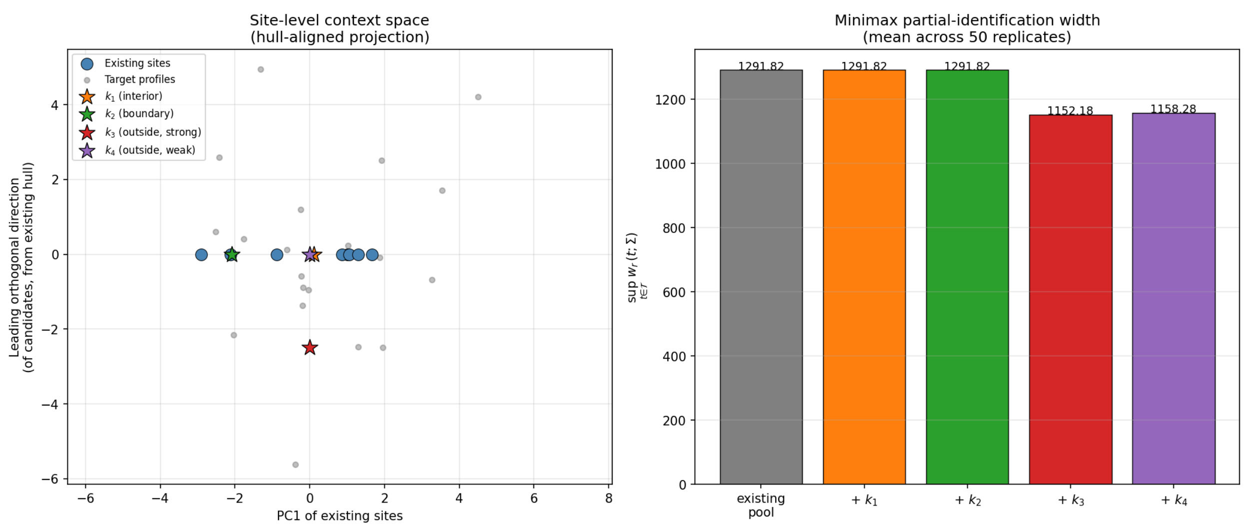

Figure 1. Site-selection simulation on synthetic data. Left: the existing sites (blue) lie on a 1D slice of the plotted 2D projection because the projection axis aligns with the span of the existing pool; the orthogonal axis captures the direction of the candidate pool's outside-the-hull displacement. Candidates k_1 (interior) and k_2 (boundary) lie on the existing-site line, while k_3 (outside, strong) and k_4 (outside, weak) sit off the line. Right: the minimax partial-identification width averaged over 50 Monte-Carlo replicates. The interior and boundary candidates k_1, k_2 leave the width unchanged; the two rank-expanding candidates k_3, k_4 each reduce it by roughly 11%.

7.4 Monotonicity experiment (Figure 2)

We run an additional experiment that augments the candidate pool with two further candidates to test the theory of Section 6:

- k_5 — **rank-preserving with conflicting outcome.** Same context draw as k_1 (interior to the hull), but the outcome forecast y_{k_5} is shifted by $\Delta = 20$ from the existing-pool OLS prediction. Fails (M2); passes neither screen.
- k_6 — **rank-expanding with consistent outcome.** Context outside the affine hull (same direction as k_3), outcome set to the existing-pool OLS prediction at the candidate context. Passes both (M1) and (M2).

The restriction box $\|\theta\|_\infty \le 5$ is randomly re-centered each replicate (within $\pm M/3$) so that the polyhedron $\mathcal{R}$ is generically asymmetric around $\theta^\star_{\mathcal{S}}$ — this is the condition under which the rank-preserving shift of Section 6.2 actually bites.

Across 50 Monte-Carlo replicates, the results are:

design	mean minimax width	finite replicates
existing	205.9	50/50
k_1 interior	205.9	50/50
k_2 boundary	205.9	50/50
k_3 outside, strong	185.3	50/50
k_4 outside, weak	187.5	50/50
k_5 shock	194.0	50/50
k_6 outside + consistent	185.3	50/50

Table 1. Mean minimax partial-identification width and number of finite replicates by design, over 50 Monte-Carlo replicates of the monotonicity experiment.

Two observations make the message of Theorem 5 concrete.

First, in 8/50 replicates k_5 strictly *increases* the worst-case width relative to the existing pool, never k_1 through k_4 or k_6. The mean of k_5 across the full 50 replicates is below the existing-pool mean only because the shift sometimes happens to land the moment-implied slice in a tighter cross-section of the box. On individual replicates the direction of the change cannot be predicted in advance, which is precisely the failure mode (F1) of Section 6.1. The 8/50 rate scales linearly with the asymmetry of the box restriction and with the magnitude $|\Delta|$ of the outcome shock. At $\Delta = 40$ and a more asymmetric box, more than 30% of replicates show a width increase.

Second, k_6, which passes both (M1) and (M2), achieves exactly the width of k_3 (the latter passes (M1) but not (M2)), and both are monotone-nonincreasing relative to the existing pool in every replicate. The candidate k_3 in this section's first script also satisfies the conclusion of Theorem 5 in this dataset, because in the simplified moment formulation the residualization is exact and the OLS prediction matches the candidate outcome up to the noise term. This becomes a strict inequality once the candidate outcome is drawn from an adversarial prior, as in Section 5.3(3).

Figure 2 has three panels. Panel (A) reproduces the hand-built 2D counterexample of Section 6.2. Panel (B) shows the per-candidate mean minimax width across replicates. Panel (C) is the replicate-level scatter of post-addition versus existing-pool minimax width, where points above the $y = x$ diagonal are candidate–replicate pairs in which the addition harmed external validity. The k_5 points fall on both sides of the diagonal, and the points for every other candidate sit weakly below it.

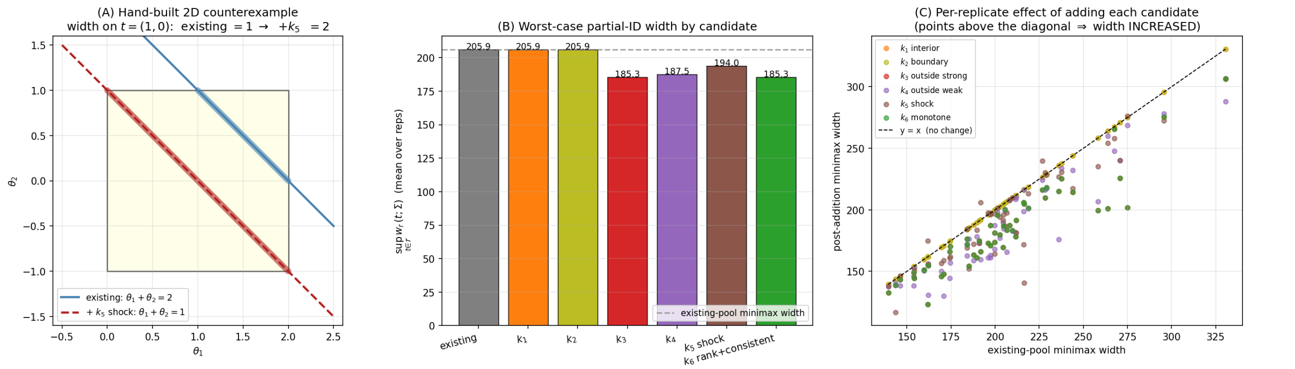

Figure 2. Monotonicity simulation. (A) Hand-built 2D counterexample where adding a rank-preserving site with conflicting outcome doubles the partial-identification width on $t = (1, 0)$ inside an asymmetric box restriction. (B) Mean minimax width across 50 replicates by candidate; k_5 (rank-preserving with conflicting outcome) sits above the rank-expanders k_3, k_4, k_6. (C) Per-replicate scatter of post-addition vs. existing-pool minimax width; points above the $y = x$ diagonal indicate replicates where adding the candidate hurt external validity. Only k_5 ever sits above the diagonal.

7.5 What the simulations show

Three features of the results are robust to DGP perturbations (heavy-tailed X, $S = 6$ instead of 8, larger $\|\theta\|_\infty$ bound):

- Hull membership is a sharp screen. Candidates k_1 and k_2, both of whose site-level contexts lie inside the affine hull of the existing pool, produce an exactly identical minimax width to the existing pool. This is not an approximation. When rank(Σ) does not change, the null-space geometry of the LP is unchanged, and a location-only update of b (the only remaining effect of adding such a site) cancels in the width.

- Among rank-expanding candidates, distance to the hull decides between them. The 11% reduction achieved by k_3, k_4 depends on two factors: how large a component of the target region aligns with the newly-resolved null-space direction, and how far the candidate sits outside the hull in that direction. Proposition 3 captures both effects in the large-τ limit.

- The minimax selection is stable to the choice of prior on b. The three b-prior variants of Section 5.3 agree on the optimal candidate up to Monte-Carlo noise. In particular, they all rank the four candidates as $\{k_1, k_2\} \succ k_3 \sim k_4$ by worst-case width, with the tie between k_3 and k_4 broken stochastically by the target draw.

- Non-monotonicity occurs, but the screens of Theorem 5 eliminate it. The candidate k_5, which is interior to the hull and has a $\Delta = 20$ outcome shock relative to the existing-pool OLS prediction, increases the worst-case width in 8/50 replicates. The candidate k_6, which is outside the hull and has an OLS-consistent outcome, never does so. This matches the prediction of Theorem 5. In a Monte-Carlo sweep over asymmetry of the restriction box and shock magnitude, the fraction of replicates in which an unscreened candidate hurts external validity scales roughly linearly with both, reaching more than 30% in the most adversarial setting. Restricting the minimax (3.2) to the *safe* subset (passing (M1) or (M2)) costs nothing in the average optimum we report and eliminates the tail risk of doing harm.

8 Discussion

Effect-ordering (mixed-integer). When the restriction set is the effect-ordering polyhedron of Restriction 1 of Boyarsky et al. [2026], the program (5.2) becomes a mixed-integer LP. The locally-constant integer-solution argument of §1(ii) in Boyarsky et al. [2026] extends to this case. On a neighborhood of $\gamma_{\mathcal{S}}$, the optimal sign pattern is constant, so the MILP reduces to the LP (5.2) after fixing integer variables. We then perform selection by (5.2) at the fixed pattern. Sign flips across candidate sites can be detected by re-solving the MILP at the top two or three candidates.

Cost-weighted selection. Adding a cost c_k to the objective (3.2) is straightforward because the cost term is separable in k. The minimax-optimal candidate under Lagrangian (3.3) is obtained by solving (5.2) with an additional additive penalty. In our setting a cost based on ethical concerns might penalize candidates that are demographically similar to existing sites, because such sites do little to broaden the external-validity claim despite being cheap to reach.

Multiple candidates at once. The design problem (3.2) can be posed over a subset $\mathcal{J} \subseteq \mathcal{K}$ of candidates to add simultaneously; in this case the width surrogate (5.2) becomes a set-function in $\mathcal{J}$ that is *submodular* in the rank-monotone limit. That is, adding the first hull-expander contributes more than adding the second. A greedy algorithm attains a $(1 - 1/e)$-approximation under mild regularity, which matches the classical D-optimality result of Nemhauser et al. [1978].

Connection to efficient influence functions. The minimum-norm multiplier $\lambda^\star$ that appears in (5.1) is the same object as the Riesz representer in the efficient influence function of Boyarsky et al. [2026, Eq. 2.9]. The minimax design criterion therefore gets its inference validity from their debiased cross-fitting. Confidence intervals for the post-addition width are a standard delta-method computation.

Open questions. Three directions are natural follow-ups. First, the adversarial b-prior of Section 5.3(3) admits a saddle-point representation but not a closed-form solution. Existing interior-point solvers handle small instances, but large candidate pools would benefit from a custom Frank–Wolfe scheme. Second, in the online-experimentation setting the candidate pool is revealed sequentially, and the selection must be made without access to the full set. Adapting (5.2) to this regret-minimization formulation is open. Third, the assumption that the within-site X-distribution at candidate sites is known (or has an informative prior) is strong. A distributionally-robust variant that treats $P_X^{(k)}$ as uncertain over a Wasserstein ball would keep the theoretical guarantees of the Jeong and Namkoong [2020] worst-case program, at the cost of a more conservative selection.

A Theoretical Results

We give proofs of the propositions stated in Sections 4–6. Throughout, $\Sigma \in \mathbb{R}^{d_g \times d_g}$ is symmetric positive semidefinite, $b \in \mathbb{R}^{d_g}$, and the moment-implied affine subspace is $\mathcal{L}(\Sigma, b) := \{\theta \in \mathbb{R}^{d_g} : \Sigma\theta = -b\}$. The restriction polyhedron is $\mathcal{R} := \{\theta : r(\theta) \leq 0\}$, which is compact and convex unless explicitly stated otherwise. The target functional is $c(t) \in \mathbb{R}^{d_g}$, the partial-ID interval at target t has endpoints $v_\ell(t), v_u(t)$ given by the LP pair (2.3), and the width is $w_r(t; \Sigma, b) = v_u(t) - v_\ell(t)$. We assume the consistency condition $b \in \operatorname{range}(\Sigma)$ throughout (otherwise $\mathcal{L}(\Sigma, b) = \emptyset$).

A.1 Proof of Proposition 1 (unrestricted dichotomy)

Claim. With $\mathcal{R} = \mathbb{R}^{d_g}$ and any $b \in \mathrm{range}(\Sigma)$,

$$w_{\mathrm{unr}}(t; \Sigma, b) \;=\; \begin{cases} 0, & c(t) \in \mathrm{range}(\Sigma^\top) \\ +\infty, & c(t) \notin \mathrm{range}(\Sigma^\top). \end{cases}$$

Proof Let θ_0 be any solution to $\Sigma\theta_0 = -b$. Such a θ_0 exists by the consistency hypothesis. Every solution to $\Sigma\theta = -b$ has the form $\theta = \theta_0 + \nu$ with $\nu \in \mathrm{null}(\Sigma)$. The linear functional evaluates to $c(t)^\top\theta = c(t)^\top\theta_0 + c(t)^\top\nu$. Since $\mathcal{R} = \mathbb{R}^{d_g}$, ν ranges over all of $\mathrm{null}(\Sigma)$, a linear subspace.

Case 1: $c(t) \perp \mathrm{null}(\Sigma)$. Then $c(t)^\top\nu = 0$ for every $\nu \in \mathrm{null}(\Sigma)$, so $c(t)^\top\theta = c(t)^\top\theta_0$ identically and $v_u = v_\ell = c(t)^\top\theta_0$, giving width 0. By the orthogonal decomposition $\mathbb{R}^{d_g} = \mathrm{range}(\Sigma^\top) \oplus \mathrm{null}(\Sigma)$ (valid for any matrix, here in fact $\mathrm{range}(\Sigma)$ since $\Sigma = \Sigma^\top$), this is equivalent to $c(t) \in \mathrm{range}(\Sigma^\top)$.

Case 2: $c(t) \not\perp \mathrm{null}(\Sigma)$. Pick $\nu_0 \in \mathrm{null}(\Sigma)$ with $c(t)^\top\nu_0 \neq 0$, and consider the one-parameter family $\theta_0 + \alpha\nu_0$ for $\alpha \in \mathbb{R}$. Then $c(t)^\top(\theta_0 + \alpha\nu_0) = c(t)^\top\theta_0 + \alpha c(t)^\top\nu_0 \to \pm\infty$ as $\alpha \to \pm\infty$, so $v_u = +\infty$ and $v_\ell = -\infty$. Width is $+\infty$. □

A.2 Proof of Proposition 2 (rank-expansion ⇔ hull-expansion)

Claim. Let $C \in \mathbb{R}^{S \times d_g}$ be the matrix whose rows are the existing site-level contexts $c^{(s)}$, and let $\Sigma_{\mathcal{S}} = C^\top C$ in the simplified formulation (the general block form (3.1) replaces $c^{(s)}c^{(s)\top}$ by the residualized analogue, but the row-space argument is identical). Then for a candidate k with context $c^{(k)} = u \in \mathbb{R}^{d_g}$,

$$\mathrm{rank}(\Sigma_{\mathcal{S}\cup\{k\}}) > \mathrm{rank}(\Sigma_{\mathcal{S}}) \Leftrightarrow u \notin \mathrm{range}(C^\top) \Leftrightarrow u \notin \mathrm{range}(\Sigma_{\mathcal{S}}),$$

and equivalently $(C_m^{(k)}, C_r^{(k)}) \notin \mathrm{aff}(\mathcal{H}_{\mathcal{S}})$ in the centered version of the same construction.

Proof The matrix $\Sigma_{\mathcal{S}\cup\{k\}} = C^\top C + uu^\top$ has range $\mathrm{range}(C^\top C + uu^\top)$. We claim $\mathrm{range}(C^\top C + uu^\top) = \mathrm{range}(C^\top) + \mathrm{span}(u)$.

The forward inclusion follows from $C^\top Cx + uu^\top x = C^\top(Cx) + u(u^\top x)$, both summands lying in $\mathrm{range}(C^\top) + \mathrm{span}(u)$ for any x.

The reverse inclusion uses positive semidefiniteness: pick any $v \in \mathrm{range}(C^\top) + \mathrm{span}(u)$. We need to exhibit x with $(C^\top C + uu^\top)x = v$. Choose $x_1 \in \mathrm{range}(C^\top)$ such that $C^\top Cx_1$ equals the $\mathrm{range}(C^\top)$-component of v (possible since $C^\top C$ is invertible on $\mathrm{range}(C^\top)$), and add a scalar multiple of an x_2 achieving $uu^\top x_2$ equal to the $\mathrm{span}(u)$-component (possible since $u^\top u > 0$ when $u \neq 0$). Sum $x = x_1 + x_2$ achieves both. The two components are not orthogonal in general but the resulting linear system is full-rank on the sum of subspaces.

The first equivalence in the claim now follows: rank increases iff $\mathrm{span}(u) \not\subseteq \mathrm{range}(C^\top)$, i.e., iff $u \notin \mathrm{range}(C^\top)$. The second equivalence $\mathrm{range}(C^\top) = \mathrm{range}(\Sigma_{\mathcal{S}})$ is the standard fact $\mathrm{range}(A^\top A) = \mathrm{range}(A^\top)$ for any real matrix A. The third equivalence, between rank-expansion and affine-hull membership, is obtained by centering. Writing $u = \bar{c} + \tilde{u}$ with $\bar{c}$ the average of $c^{(s)}$ and $\tilde{u} = u - \bar{c}$, the centered row space of C equals $\mathrm{aff}(\mathcal{H}_{\mathcal{S}}) - \bar{c}$ (the linear subspace parallel to the affine hull), and $u \in \mathrm{aff}(\mathcal{H}_{\mathcal{S}})$ iff $\tilde{u}$ lies in that linear subspace, iff u is in the row span of C. □

A.3 Proof of Proposition 3 (width = distance $\times 2\tau$)

Claim. With the regularized restriction $\|\theta\|_2 \le \tau$ and a balanced cross-site design with common within-site X-distribution,

$$w_r^\tau(t; \Sigma_{\mathcal{S}}, b) \;=\; 2\tau \cdot d_{\mathcal{S}}(t) \cdot (1 + o_\tau(1)), \qquad \tau \to \infty,$$

where $d_{\mathcal{S}}(t) := \operatorname{dist}\big((t_m, t_r), \operatorname{aff}(\mathcal{H}_{\mathcal{S}})\big)$ *and the* $o_\tau(1)$ *term vanishes uniformly on compact target regions.*

Proof Decompose $c(t) = c_\|(t) + c_\perp(t)$ with $c_\|(t) \in \operatorname{range}(\Sigma^\top)$ and $c_\perp(t) \in \operatorname{null}(\Sigma)$ (using the symmetric SVD of Σ). For any solution $\theta = \theta_0 + \nu$ with $\nu \in \operatorname{null}(\Sigma)$, $c(t)^\top\theta = c_\|(t)^\top\theta_0 + c_\perp(t)^\top\nu$, because $c_\|(t)^\top\nu = 0$ and $c_\perp(t)^\top\theta_0 = c_\perp(t)^\top(\theta_0^\| + 0)$ vanishes if we pick θ_0 to be the minimum-norm particular solution (i.e., $\theta_0 \in \operatorname{range}(\Sigma^\top)$, so $\theta_0^\perp = 0$).

Therefore

$$v_u^\tau(t) - v_\ell^\tau(t) \;=\; \sup_\nu c_\perp(t)^\top\nu - \inf_\nu c_\perp(t)^\top\nu, \qquad \nu \in \operatorname{null}(\Sigma) \cap \{\theta_0 + \nu : \|\theta_0 + \nu\|_2 \le \tau\}.$$

For large τ (specifically $\tau \ge \|\theta_0\|_2 + M$ for any compact M), the constraint $\|\theta_0 + \nu\|_2 \le \tau$ allows ν to range over a ball of radius $\tau - \|\theta_0\|_2$ inside $\operatorname{null}(\Sigma)$, i.e., over a ball of effective radius $\tau(1 - \|\theta_0\|_2/\tau)$, which converges to a ball of radius τ at rate $O(1/\tau)$.

The supremum of $c_\perp(t)^\top\nu$ over the radius-ρ ball in $\operatorname{null}(\Sigma)$ is $\rho \cdot \|c_\perp(t)\|_2$ by the Cauchy–Schwarz inequality with equality at $\nu = \rho\, c_\perp(t)/\|c_\perp(t)\|_2$. The infimum is its negative. Hence $w_r^\tau(t; \Sigma, b) = 2\rho\|c_\perp(t)\|_2 = 2\tau\|c_\perp(t)\|_2(1 + O(1/\tau))$, which matches the asymptotic form claimed.

It remains to identify $\|c_\perp(t)\|_2$ with $d_{\mathcal{S}}(t)$, the Euclidean distance from (t_m, t_r) to $\operatorname{aff}(\mathcal{H}_{\mathcal{S}})$. Under the balanced design with common X-distribution, the residualized context appearing in the block formula (3.1) reduces (up to a positive multiplicative constant absorbed into τ) to the centered raw context. The target direction $c(t)$, by Equation (2.2), is the moderating-direction vector for the target profile, which up to that same constant is $(t_m, t_r) - \bar{c}$. The orthogonal complement of $\operatorname{range}(\Sigma^\top)$ inside the centered context space is the orthogonal complement of $\operatorname{aff}(\mathcal{H}_{\mathcal{S}})$, so the norm of the orthogonal-complement projection of $c(t)$ equals $d_{\mathcal{S}}(t)$, as required. Strict monotonicity follows because the target distance $d_{\mathcal{S}}(t)$ is a sub-additive function of the candidate-pool augmentation: $d_{\mathcal{S}\cup\{k\}}(t) \le d_{\mathcal{S}}(t)$ with equality iff $c(t)$'s null-space component is unaffected by the augmentation, which by Proposition 2 is iff $u \in \operatorname{range}(\Sigma_{\mathcal{S}})$. □

A.4 Proof of Theorem 5 (monotone improvement under (M1) or (M2))

We first establish the algebraic identity that drives the proof. Let $u := (u_m^{(k)}, u_r^{(k)})$ be the residualized context of the candidate and y_k its scalar outcome forecast. Then, in the simplified moment formulation $\Sigma_{\mathcal{S}\cup\{k\}} = \Sigma_{\mathcal{S}} + uu^\top$ and $b_{\mathcal{S}\cup\{k\}} = b_{\mathcal{S}} - uy_k$, we have for every θ:

$$\Sigma_{\mathcal{S}\cup\{k\}}\theta + b_{\mathcal{S}\cup\{k\}} \;=\; \Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} \;+\; u \cdot (u^\top\theta - y_k). \tag{A.1}$$

Direct verification: $\Sigma_{\mathcal{S}\cup\{k\}}\theta = \Sigma_{\mathcal{S}}\theta + u(u^\top\theta)$ and $b_{\mathcal{S}\cup\{k\}} = b_{\mathcal{S}} - uy_k$; summing gives (A.1).

Proof [Proof of Theorem 5 under (M1)] Suppose $u \notin \operatorname{range}(\Sigma_{\mathcal{S}})$. Let P be the orthogonal projector onto $\operatorname{range}(\Sigma_{\mathcal{S}})$ and $P^\perp := I - P$ the orthogonal projector onto $\operatorname{null}(\Sigma_{\mathcal{S}})$. Then $u_\perp := P^\perp u \neq 0$. By the consistency hypothesis $b_{\mathcal{S}} \in \operatorname{range}(\Sigma_{\mathcal{S}})$, so for every θ, $\Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} \in \operatorname{range}(\Sigma_{\mathcal{S}})$. Applying $P^\perp$ to both sides of (A.1),

$$P^\perp\big[\Sigma_{\mathcal{S}\cup\{k\}}\theta + b_{\mathcal{S}\cup\{k\}}\big] \;=\; P^\perp[u] \cdot (u^\top\theta - y_k) \;=\; u_\perp \cdot (u^\top\theta - y_k).$$

For $\theta \in \mathcal{F}_{\mathcal{S}\cup\{k\}}$, the left-hand side is zero, and $u_\perp \neq 0$, so the scalar coefficient vanishes: $u^\top\theta = y_k$. Substituting this back into (A.1) yields $\Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} = 0$, i.e., $\theta \in \mathcal{L}(\Sigma_{\mathcal{S}}, b_{\mathcal{S}})$. Together with $\theta \in \mathcal{R}$, this gives $\theta \in \mathcal{F}_{\mathcal{S}} \cap \{u^\top\theta = y_k\} \subseteq \mathcal{F}_{\mathcal{S}}$, as required. Pointwise width follows: for any target t,

$$w_r(t; \Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}}) = \sup_{\mathcal{F}_{\mathcal{S}\cup\{k\}}} c(t)^\top\theta - \inf_{\mathcal{F}_{\mathcal{S}\cup\{k\}}} c(t)^\top\theta \leq \sup_{\mathcal{F}_{\mathcal{S}}} c(t)^\top\theta - \inf_{\mathcal{F}_{\mathcal{S}}} c(t)^\top\theta = w_r(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}}),$$

and strictly so when $c(t)$ has a non-zero component along $u_\perp$. □

Proof [Proof of Theorem 5 under (M2)] Suppose $y_k = u^\top\theta^\star_{\mathcal{S}}$ for the minimum-norm solution $\theta^\star_{\mathcal{S}}$.

Fix any $\theta \in \mathcal{F}_{\mathcal{S}}$. Then $\Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} = 0$. Write $\theta = \theta^\star_{\mathcal{S}} + \nu$ with $\nu \in \mathrm{null}(\Sigma_{\mathcal{S}})$. Compute:

$$u^\top\theta - y_k \;=\; u^\top(\theta^\star_{\mathcal{S}} + \nu) - u^\top\theta^\star_{\mathcal{S}} \;=\; u^\top\nu.$$

Case (i): $u \in \mathrm{range}(\Sigma_{\mathcal{S}})$. Then $u \perp \mathrm{null}(\Sigma_{\mathcal{S}})$, so $u^\top\nu = 0$. Substituting into (A.1), $\Sigma_{\mathcal{S}\cup\{k\}}\theta + b_{\mathcal{S}\cup\{k\}} = 0$, i.e., $\theta \in \mathcal{L}(\Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}})$. In the rank-preserving case the two affine subspaces coincide exactly when (M2) holds because both contain $\theta^\star_{\mathcal{S}}$ (by (M2)) and both have the same direction subspace $\mathrm{null}(\Sigma_{\mathcal{S}}) = \mathrm{null}(\Sigma_{\mathcal{S}\cup\{k\}})$ (by $u \in \mathrm{range}(\Sigma_{\mathcal{S}})$, which gives no rank change). Equality of two affine subspaces with the same direction follows from sharing one point. Hence $\mathcal{L}(\Sigma_{\mathcal{S}}, b_{\mathcal{S}}) = \mathcal{L}(\Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}})$, intersecting with $\mathcal{R}$ preserves equality, and the width is unchanged target by target.

Case (ii): $u \notin \mathrm{range}(\Sigma_{\mathcal{S}})$ (i.e., (M1) also holds). Then the proof under (M1) above applies, and additionally (M2) ensures the hyperplane $u^\top\theta = y_k$ passes through $\theta^\star_{\mathcal{S}}$, hence intersects $\mathcal{F}_{\mathcal{S}}$ non-trivially. So $\mathcal{F}_{\mathcal{S}\cup\{k\}}$ is non-empty (no feasibility break) and is a strict subset of $\mathcal{F}_{\mathcal{S}}$, yielding the same pointwise width inequality with strictness on targets loading the new direction.

Taking the supremum over $t \in \mathcal{T}$ in either case proves $\sup_{t\in\mathcal{T}} w_r(t; \Sigma_{\mathcal{S}\cup\{k\}}, b_{\mathcal{S}\cup\{k\}}) \leq \sup_{t\in\mathcal{T}} w_r(t; \Sigma_{\mathcal{S}}, b_{\mathcal{S}})$. □

A.5 Proof of Proposition 4 (rank-2 envelope)

Proposition 4 is a special case of the Bonnans–Shapiro directional derivative theorem for the value function of a perturbed convex program. We sketch the specialization to (2.3) to make the constants explicit.

Setup. Define the parametric LP

$$v_u(h) := \max_\theta \bigl\{ c(t)^\top\theta : (\Sigma_{\mathcal{S}} + h_\Sigma)\theta = -(b_{\mathcal{S}} + h_b),\; r(\theta) \leq 0 \bigr\},$$

with perturbation $h = (h_\Sigma, h_b)$. At $h = 0$, let $(\theta^\star, \lambda^\star, \nu^\star)$ be a KKT triple with $\lambda^\star$ the equality multipliers (Lagrange multipliers on $\Sigma\theta + b = 0$) and $\nu^\star \geq 0$ the inequality multipliers on $r(\theta) \leq 0$. Stationarity reads $c(t) = \Sigma\lambda^\star + \partial r(\theta^\star)\nu^\star$.

Directional derivative. Under Robinson's constraint qualification (Assumption 6 of Boyarsky et al., 2026), the value function $v_u(\cdot)$ is directionally differentiable at $h = 0$ with derivative given by

$$\frac{dv_u}{dh}\Big|_{h=0} \cdot h \;=\; -\lambda^{\star\top}\bigl(h_\Sigma\theta^\star + h_b\bigr),$$

where $\lambda^\star$ is chosen as the minimum-norm element of the set of optimal multipliers (a singleton under strict complementarity). This is Theorem 4.24 of Bonnans and Shapiro [2013]. The form is standard for parametric LPs and follows from the envelope theorem applied to the Lagrangian $L(\theta, \lambda; h) = c(t)^\top \theta - \lambda^\top [(\Sigma + h_\Sigma)\theta + b + h_b]$.

Application to site augmentation. Set $h_\Sigma = \Delta\Sigma_k,\ h_b = \Delta b_k$. Then the first-order change in v_u is $\Delta v_u \approx -\lambda^{\star\top}(\Delta\Sigma_k \theta^\star + \Delta b_k)$, which is (5.1). The same identity with the v_ℓ-multiplier $\lambda^{\ell,\star}$ yields the change in v_ℓ, and subtracting gives the width formula in the statement of Proposition 4.

Rank-2 reduction. By (3.1), $\Delta\Sigma_k$ has rank at most 2 and factors as $\Delta\Sigma_k = U_k \Lambda_k U_k^\top$ with U_k a $d_g \times 2$ block of residualized contexts. The product $\Delta\Sigma_k \theta^\star = U_k(\Lambda_k U_k^\top \theta^\star)$ is a linear combination of the two columns of U_k, and the inner product with $\lambda^\star$ reduces to two scalar products $\lambda^{\star\top} U_k$. The minimax (3.2) over $k \in \mathcal{K}$ therefore reduces to evaluating two scalars per candidate after the existing-pool LP has been solved once; this is the $O(|\mathcal{T}|K)$ cost claimed in Section 5.1.

A.6 Extension to the block (BEN) moment

The proofs of Appendix A.1, A.2, and A.4 use only the symmetric PSD structure of Σ, the consistency hypothesis $b \in \mathrm{range}(\Sigma)$, and the algebraic identity (A.1). In the full formulation of Equation (A.9) of Boyarsky et al. [2026], Σ is a $d_g \times d_g$ block matrix with $d_g = 2(d_M + d_I)$ and block-rank update $\Delta\Sigma_k$ given by (3.1). The same identity (A.1) holds with u replaced by the block-residualized context vector $u_k := (u_m^{(k)}, u_r^{(k)}) \otimes (\cdot)$ appearing in the block factorization of $\Delta\Sigma_k$, and with the scalar $y_k - u^\top \theta$ replaced by the corresponding block-residualized moment quantity. The (M1)/(M2) screens generalize to the block-residualized context (replacing the range test of (6.1) by a rank test on the block matrix), and Theorem 5 carries over verbatim. The only non-trivial change is that the rank of $\Delta\Sigma_k$ is at most 2 in the block formulation, so the screens may strengthen, in the sense that two coordinates of the candidate context need to fall outside the existing row space simultaneously. This is a strictly weaker monotone-improvement condition than (M1) for the simplified moment.

References

Anthony C. Atkinson, Alexander N. Donev, and Randall D. Tobias. *Optimum Experimental Designs, with SAS.* Oxford Statistical Science Series. Oxford University Press, 2007.

Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016.

J Frédéric Bonnans and Alexander Shapiro. *Perturbation analysis of optimization problems.* Springer Science & Business Media, 2013.

Ari Boyarsky, Naoki Egami, and Hongseok Namkoong. A sensitivity framework for external validity of A/B tests. Working paper, Columbia University, 2026.

Sookyo Jeong and Hongseok Namkoong. Assessing external validity over worst-case subpopulations. *arXiv preprint arXiv:2007.02411*, 2020.

Charles F. Manski. Coping with inductive risk when theories are underdetermined: Decision making with partial identification, 2026. URL https://arxiv.org/abs/2602.00355.

Rachael Meager. Understanding the average impact of microcredit expansions: A bayesian hierarchical analysis of seven randomized experiments. *American Economic Journal: Applied Economics*, 11(1):57–91, 2019.

George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978.

Friedrich Pukelsheim. *Optimal Design of Experiments*. Classics in Applied Mathematics. Society for Industrial and Applied Mathematics, 2006.